

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Artificial Intelligence-Driven X-Ray Analysis: Enhancing Image Quality and Diagnostic Accuracy

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- Chest X-ray images are an essential component of contemporary medicine.
- They are widely used because they are low-cost and easily available.
- Radiologists face challenges in interpreting chest X-rays due to subtle abnormalities and image quality variations.
- Fatigue, human error, and delays may affect diagnosis.
- Artificial Intelligence helps radiologists improve consistency and diagnostic accuracy.
- AI systems can assist in fast diagnosis and efficient screening of chest-related diseases.

- Manual interpretation of chest X-rays is difficult.
- Image quality and contrast differences create diagnostic challenges.
- Human workload may reduce accuracy in diagnosis.
- Rural and isolated areas have limited access to radiologists.
- There is a need for intelligent systems that support faster and more accurate diagnosis.

- **Enhancement**

Improve the quality of chest X-ray images.

- **Detection**

Develop a strong classification model for abnormalities.

- **Explainability**

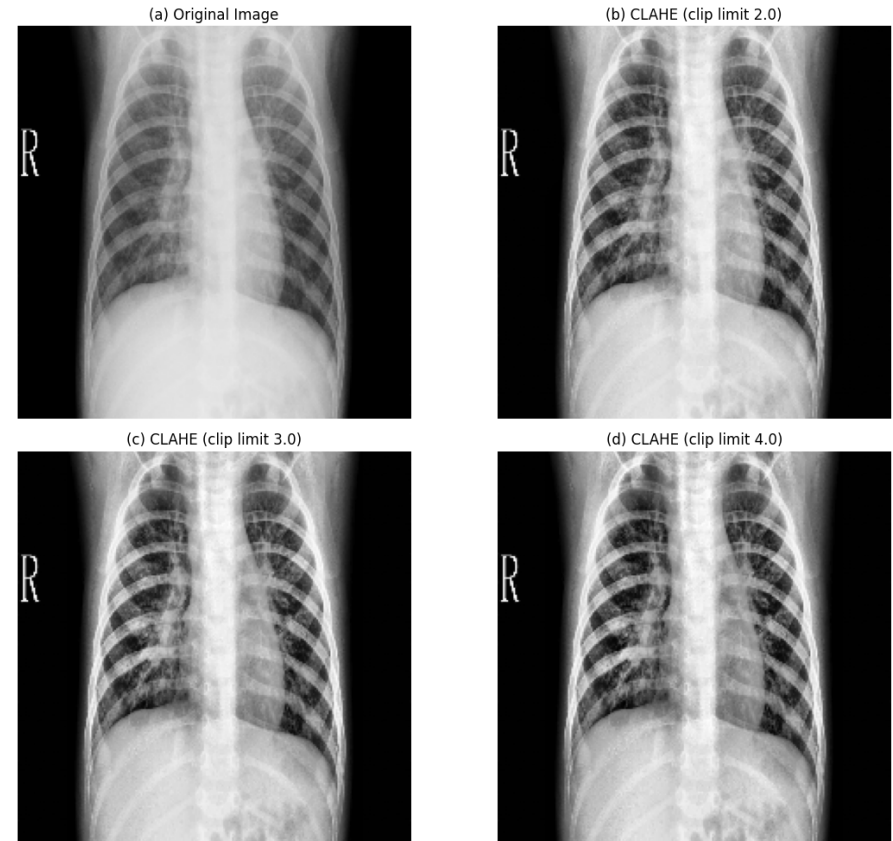
Use explainable AI to improve transparency and trust.

- CNN architectures such as VGG, ResNet, and DenseNet are used for disease detection.
- Transfer learning helps achieve good results with small medical datasets.
- CLAHE improves image contrast and feature visibility.
- Grad-CAM improves transparency of AI-based predictive models.
- Hybrid models combining preprocessing and transfer learning improve performance.

Dataset and Preprocessing



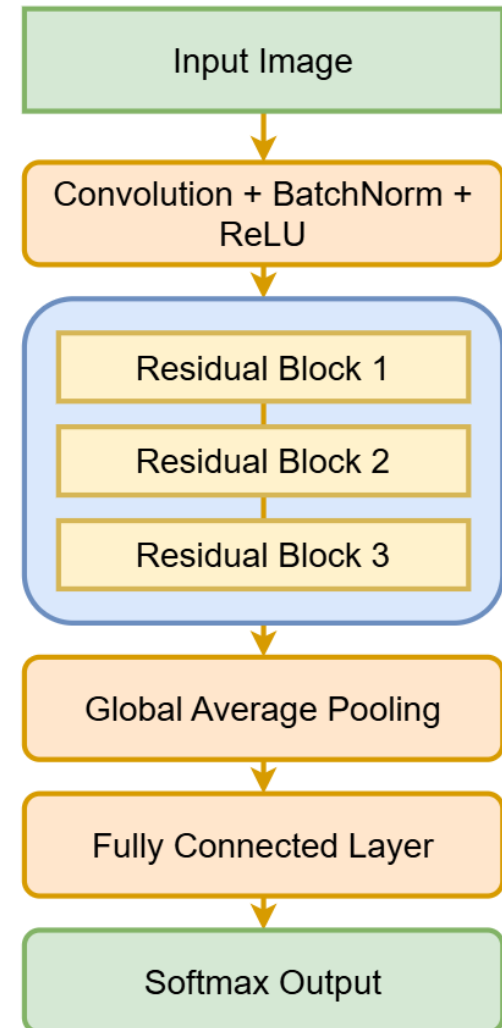
- Gaussian Blur applied for noise reduction.
- CLAHE used to enhance contrast.
- Edge sharpening used for detail enhancement.
- Images resized to 224×224 and normalized.



Model Architecture



- ResNet-18 architecture used in the study.
- Transfer learning applied instead of training from scratch.
- Final layer replaced for binary classification.
- Model classifies images into normal and abnormal categories.
- PyTorch framework used for implementation.



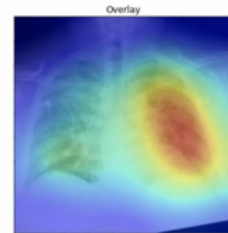
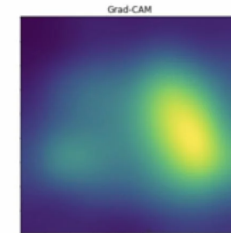
- Adam optimizer for model optimization.
- Binary cross-entropy as the loss function.
- Data augmentation (rotation/flipping).
- Early stopping and checkpointing used.
- Minimized loss via Backpropagation.
- Dataset split: Train, Val, Test.

Grad-CAM Explainability

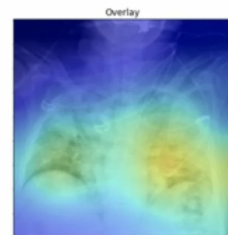
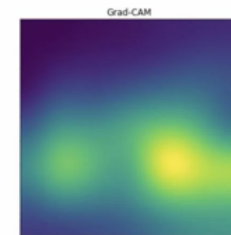


- AI models generally act as a “Black Box”.
- Grad-CAM determines critical image regions for classification.
- Heatmaps generated to explain predictions visually.
- Identifies clinically meaningful lung regions.
- Improves radiologists’ confidence in AI decisions.

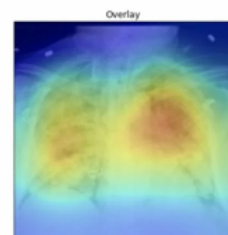
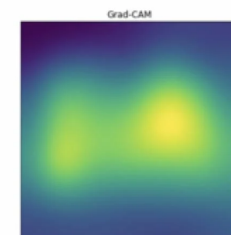
A) COVID Positive



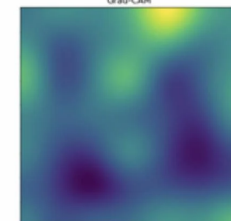
B) COVID Positive



C) COVID Positive



D) COVID Negative



Accuracy

Percentage of correct predictions.

Precision

Proportion of true positives in predicted positives.

Recall

Ability to find all positive instances.

F1 Score

Harmonic mean of Precision and Recall.

ROC-AUC

Model discrimination capability.

Results and Findings

90.1%
ACCURACY

89%
PRECISION

97%
RECALL

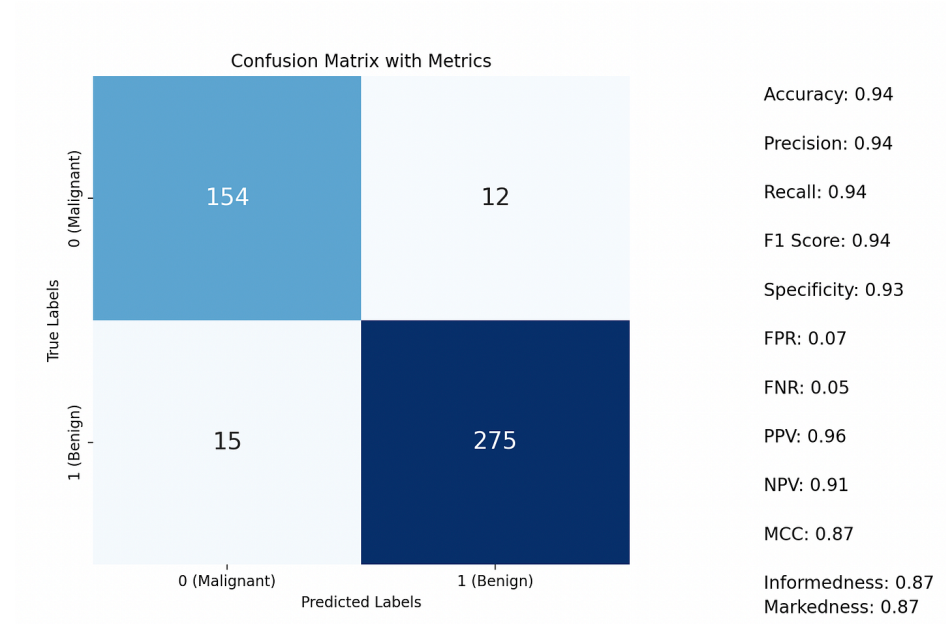
93%
F1 Score

- High recall indicates effective detection of disease cases.
- Image preprocessing significantly improved classification performance.
- Grad-CAM improved interpretability and clinical transparency.

Confusion Matrix



- Model successfully distinguished normal and abnormal chest X-rays.
- False negatives were minimized (Crucial for medical safety).
- Some false positives were observed but manageable.
- High interpretability makes the model clinically meaningful



ROC and Precision Recall



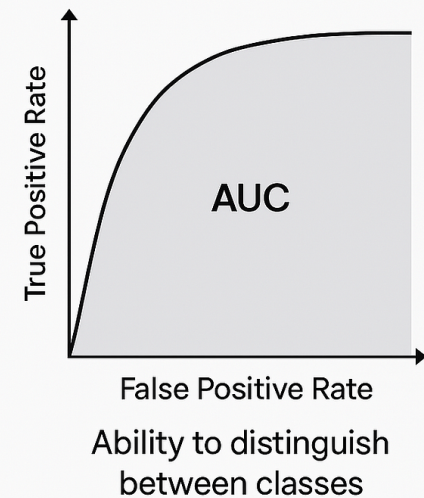
- ROC Curve shows strong discrimination ability.
- High AUC indicates reliable performance across thresholds.
- PR curve confirms stable predictive performance.
- Model performed well even on imbalanced datasets.

Confusion Matrix

	Predicted	
	True Positive (TP)	False Positive (FP)
Actual	True Positive (TP)	False Positive (FP)
	False Negative (FN)	True Negative (TN)

Performance based on
predicted vs actual
outcomes

AUC-ROC Curve



Model Comparison



Model	Accuracy	Recall	F1 Score
CNN	85%	88%	86%
VGG	89%	91%	90%
DenseNet	90%	93%	91%
Proposed Model [Hybrid]	91.5%	96.5%	92.6%

Limitations

- Small dataset size risk.
- Occasional false positives.
- High computational cost.

Future Scope

- Multi-disease detection.
- Real-time clinical support.
- Scaling with larger datasets.

- Created a strong AI-based diagnostic system for chest X-rays.
- Gaussian filtering, CLAHE, and sharpening significantly improved image quality.
- Fine-tuned ResNet-18 achieved excellent 91.5% accuracy.
- Grad-CAM provided crucial visual explanations for model predictions.
- Strong potential for integration into real-world clinical applications.



Thank You